during AI responses. You should notice that the application behaves identically to the previous version but with the added benefit of a more centralized and manageable state.

THE TAKEAWAYS

Good state management is key to building responsive, efficient, and user-friendly AI interfaces. Continually refining the state management approach will pay dividends as your application grows in complexity. You will next see some practical use cases of this system, starting with structured data generation.

Structured data generation involves prompting AI models to produce outputs in specific formats, which can greatly enhance feature development. For example, you might instruct an AI model to generate responses in formats such as JSON, Markdown for formatted text, or XML. These formats enable more organized and actionable responses, like creating detailed reports, generating interactive UI elements, or structuring data for further analysis.

4.3 Structured data generation using the Vercel AI SDK

Asking the AI model to generate text for us based on a prompt is a very useful feature, and in most cases, it's sufficient. However, AI models can do better. Why don't we ask the AI model to generate structured, type-safe data directly? After all, we might need to provide their data in another tool as input, so it makes sense to skip the part when we parse the text response and try to manually overcome this limitation.

While generating text is useful, AI models can do more. We can instruct them to generate structured, type-safe data like JSON directly, which is crucial for integrating AI responses into other applications or tools. The Vercel AI SDK provides helpful tools to do this reliably.

In this section, we'll explore how structured object generation works and provide practical examples. We'll then enhance our Astra AI agent to deliver JSON responses, which we can easily display in a React component. This approach opens new possibilities for integrating AI-generated data into various tools and applications.

4.3.1 How structured data generation works

Structured data generation is a process that allows AI models to produce organized, type-safe data instead of raw text. This approach enhances the reliability and usability of AI-generated content in various applications. Let's break down the process by exploring the diagram in figure 4.11.

As shown in the figure, the user initiates an action in a React component, and then the component calls a generation function via the Vercel AI SDK. The allowed schema is predefined by the developer when they call any of the generator functions. Then the SDK sends a structured prompt to the AI model, which in turn returns a raw response. The SDK validates the response using a schema validator, and then the structured data is returned to the client to update the UI.